

Task 2: Data Retrieval & Sorting Dashboard

Aim: To design and develop a dashboard that retrieves student records from a database, allows sorting by name or date, filtering by department, and displays the count of students in each department using SQL queries.

Algorithm:

1. Start the program
2. Connect to the database
3. Get input for sorting (name or date)
4. Get input for filtering (department)
5. Construct SQL query using sorting and filtering conditions
6. Retrieve data from the database
7. Display records in table format
8. Count students in each department using GROUP BY
9. Display count result
10. Stop the program

Program:

Db.php:

```
<?php
$conn = new mysqli("localhost", "root", "", "student_db");
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

dashboard.php:

```
<?php
include 'db.php';

// Sorting
$order = "";

if (isset($_GET['sort'])) {
    if ($_GET['sort'] == "name") {
```

```

        $order = "ORDER BY name ASC";
    } elseif ($_GET['sort'] == "dob") {
        $order = "ORDER BY dob ASC";
    }
}

// Filtering
$where = "";

if (isset($_GET['department']) && $_GET['department'] != "") {
    $dept = $_GET['department'];
    $where = "WHERE department='$dept'";
}

// Fetch student data
$sql = "SELECT * FROM students $where $order";
$result = $conn->query($sql);

// Count per department
$count_sql = "SELECT department, COUNT(*) as total FROM students GROUP BY
department";
$count_result = $conn->query($count_sql);

?>
<!DOCTYPE html>
<html>
<head>
    <title>Student Dashboard</title>
    <style>
        body {
            font-family: Arial;
            background-color: #f2f2f2;
        }
        .container {

```

```

        width: 80%;
        margin: auto;
        background: white;
        padding: 20px;
    }

    table {
        width: 100%;
        border-collapse: collapse;
        margin-top: 20px;
    }

    th, td {
        padding: 10px;
        border: 1px solid black;
    }

    button, select {
        padding: 8px;
        margin: 5px;
    }
</style>
</head>
<body>
<div class="container">
    <h2>Student Dashboard</h2>

    <!-- Sorting Buttons -->

    <a href="?sort=name"><button>Sort by Name</button></a>

    <a href="?sort=dob"><button>Sort by DOB</button></a>

    <!-- Filter Form -->

    <form method="GET">
        <select name="department">

```

```

        <option value="">All Departments</option>
        <option>CSE</option>
        <option>ECE</option>
        <option>EEE</option>
        <option>MECH</option>
    </select>

    <button type="submit">Filter</button>
</form>

<!-- Student Table -->
<table>
    <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Email</th>
        <th>DOB</th>
        <th>Department</th>
        <th>Phone</th>
    </tr>

    <?php while($row = $result->fetch_assoc()) { ?>
    <tr>
        <td><?php echo $row['id']; ?></td>
        <td><?php echo $row['name']; ?></td>
        <td><?php echo $row['email']; ?></td>
        <td><?php echo $row['dob']; ?></td>
        <td><?php echo $row['department']; ?></td>
        <td><?php echo $row['phone']; ?></td>
    </tr>

    <?php } ?>
</table>

```

```

<!-- Department Count -->
<h3>Department-wise Student Count</h3>
<table>
    <tr>
        <th>Department</th>
        <th>Total Students</th>
    </tr>
    <?php while($row = $count_result->fetch_assoc()) { ?>
        <tr>
            <td><?php echo $row['department']; ?></td>
            <td><?php echo $row['total']; ?></td>
        </tr>
    <?php } ?>
</table>
</div>
</body>
</html>

```

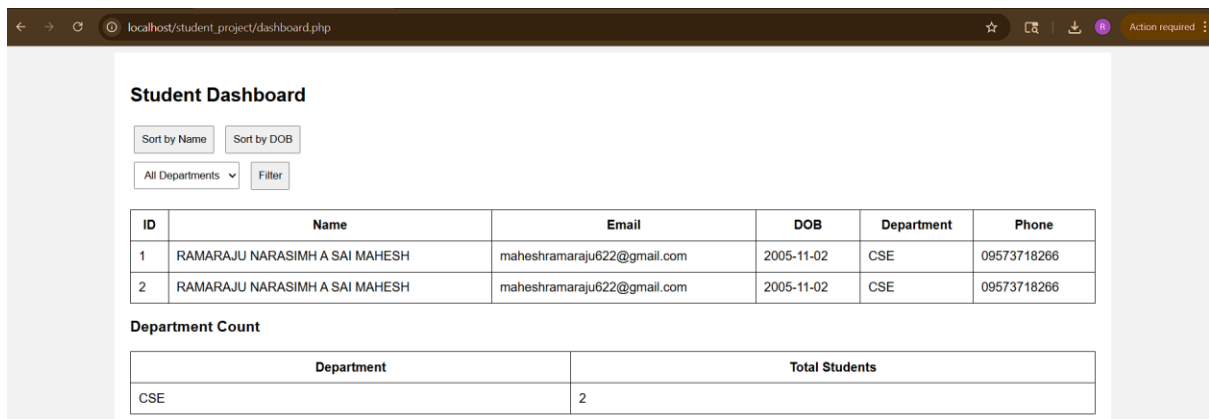
Database:

```

CREATE DATABASE student_db;
USE student_db;
CREATE TABLE students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100),
    dob DATE,
    department VARCHAR(50),
    phone VARCHAR(15)
);

```

OUTPUT:



The screenshot displays a web browser window with the address bar showing 'localhost/student_project/dashboard.php'. The page title is 'Student Dashboard'. Below the title, there are two buttons: 'Sort by Name' and 'Sort by DOB'. A dropdown menu is set to 'All Departments' with a 'Filter' button next to it. The main content area features a table with student information:

ID	Name	Email	DOB	Department	Phone
1	RAMARAJU NARASIMH A SAI MAHESH	maeshramaraju622@gmail.com	2005-11-02	CSE	09573718266
2	RAMARAJU NARASIMH A SAI MAHESH	maeshramaraju622@gmail.com	2005-11-02	CSE	09573718266

Below the student table, there is a section titled 'Department Count' with a table showing the count of students per department:

Department	Total Students
CSE	2

Result: Thus, student dashboard has been created successfully and output is generated.